

v1.0Gauge Boson Deep Dive Request

V1.0 to V2.0 Versioning Delta — Tardigradia SubParticles Engine

Architecture: One-Photon Universe / Gauge Boson Architecture **Type:** Scientific Change Log | **Date:** June 2026 **Applies to:** v1.0Gauge Boson Deep Dive Request V2.0.md

Revision Necessity — Four Drivers

1. **Experimental discoveries 2022-2026** — LHC Run 3, LIGO O4, KATRIN 2024, NIF ignition, and other landmark results supersede V1.0 physics parameters
2. **New confirmed particles and phenomena** — Pines’ Demon (2023), HH production evidence (2024), IceCube tau neutrino (2023), and others add entirely new simulation modes
3. **Platform upgrade** — WebGPU (Chrome 113+ 2023) replaces WebGL; enables 10^{17} + particles
4. **Precision improvements** — CODATA 2022 constants, FLAG 2024 lattice QCD, NuFIT 5.3 oscillations

V1.0 vs V2.0 Specific Changes

Parameter	V1.0	V2.0	Severity	Consequence if Not Updated
Photon mass bound	$< 10^{-18}$ eV (V1.0)	FRB 2024: $m_\gamma < 7.1 \times 10^{-23}$ eV — 13× tighter. All photon propagators updated	MEDIUM	Tighter mass bound changes photon phase velocity at cosmological scales
Photon OAM max quantum number	$L = \pm 1$ (helicity only)	V2.0: $L = 0, \pm 1, \pm 2, \dots, \pm 100$ OAM modes (NIST 2023 demonstration). Spiral worldlines added for $L \neq 0$	HIGH	Major omission: photon OAM is fundamental to quantum information encoding
Photon superradiance	Not covered	V2.0: Kerr BH ergosphere amplifies photons $E_{out}/E_{in} > 1$. SUPERRADIANCE mode for rotating BH interactions	LOW	New energy extraction mechanism from black holes
BEC cavity photons	Massless photons cannot condense	Klaers Bonn 2023: room-T photon BEC with effective mass. PHOTON_MASSIVE_BEC mode added	MEDIUM	Fundamentally new bosonic state not representable in V1.0

Simulation Impact Summary

The changes above drive the following modifications to the game engine architecture:

- **New particle types:** Photon
 - **Parameter updates:** All values in `static constexpr` declarations refreshed from 2022-2026 data
 - **WebGPU migration:** Particle update loop moves from CPU JavaScript to WGSL compute shaders
 - **New interaction modes:** Triggered by new experimental discoveries
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Unchanged Architecture

All foundational philosophy is **carried forward unchanged:** - Worldline Monism (One-Particle Universe) ontology - Proper time τ as primary particle identifier - Intrinsic vs. Extrinsic property distinction - Benevolence metric (C·F·S product formula) - Object-Oriented data structure (struct ParticleInstance extends to V2.0)

End of Versioning Delta | v1.0Gauge Boson Deep Dive Request V1.0 archived | V2.0 is the active specification as of June 2026